6.2. Algebra - Refresher

The Building Blocks

(x) Variables

Think of variables as containers that hold data.

In Machine Learning, we rarely work with single numbers. Instead, we represent entire datasets using symbols.

We commonly use:

- x for input features
- y for output labels

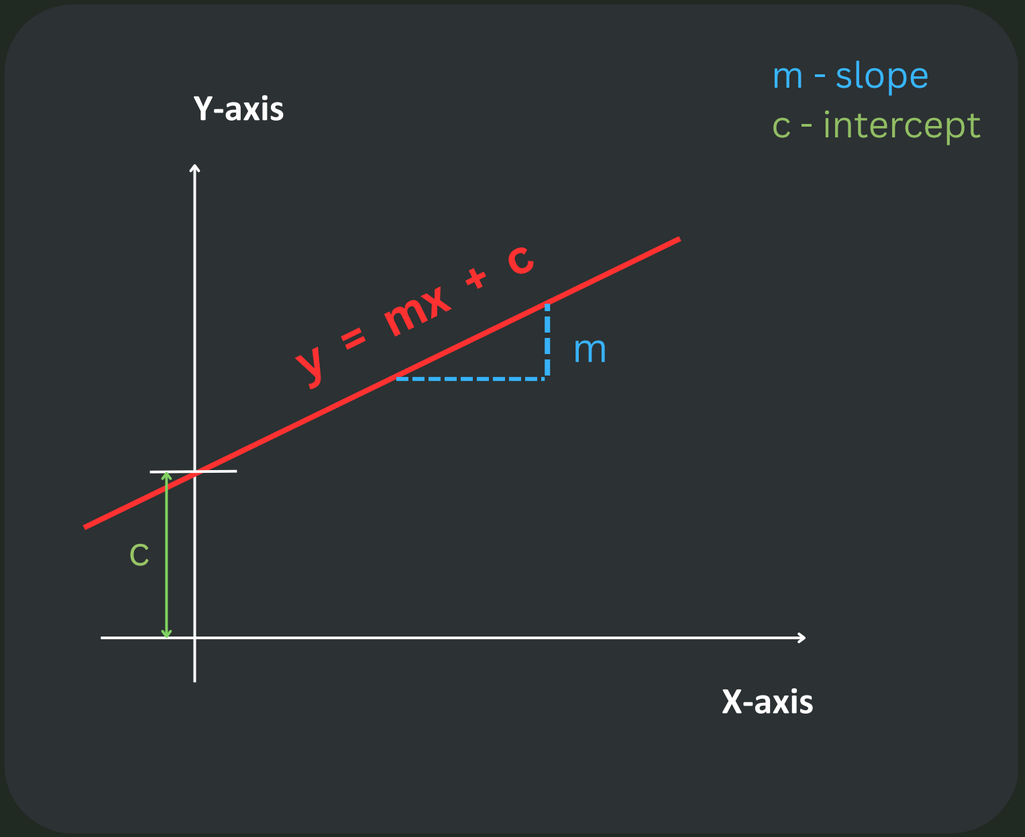
$Y = MX + C$ Equations

Equations describe how inputs relate to outputs. They are the rules a model uses to make predictions.

A Machine Learning model is essentially an equation whose parameters we adjust during training.

The goal is simple:
find the equation that best fits the data.

The Linear Function

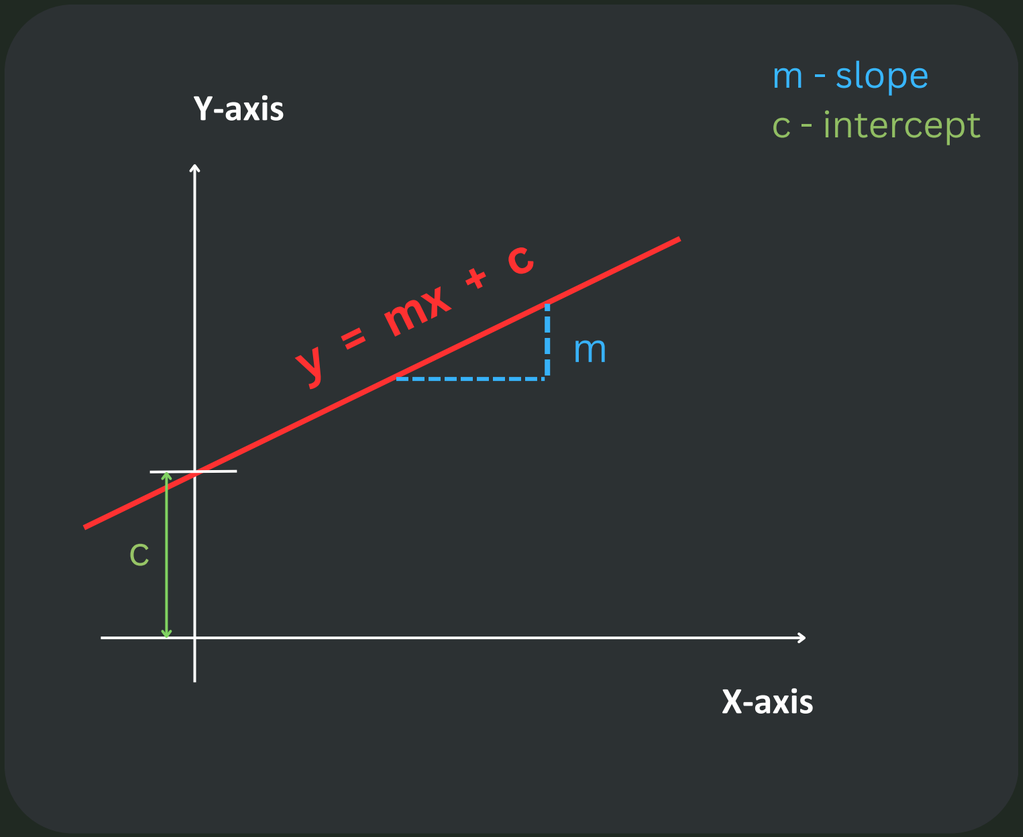


The equation of a straight line is the "Hello World" of Machine Learning algebra.

$$y = mx + c$$

- y: The output (Dependent Variable)
- x: The input (Independent Variable)
- m: Slope (Gradient) - How steep the line is.
- c: Intercept - Where the line crosses the Y-axis.

Slope & Intercept Intuition



The Slope (m)

Controls the rate of change. A positive slope means as x increases, y increases. A steep slope means a small change in input creates a large change in output.

The Intercept (c)

Controls the starting point. It effectively shifts the entire line up or down without changing its angle. It allows the model to output a non-zero value even when the input is zero.

Why This Matters - Linear Regression

Fitting the Line

Linear Regression is essentially finding the "best" slope (m) and intercept (c) so that the line passes as close as possible to all your data points.

The Core Task

In algebra, you are often given m and c and asked to plot the line. In Machine Learning, you are given the data points (dots) and asked to find the best m and c .

Translation: Algebra vs. ML

We use the exact same math, but the terminology changes slightly in the ML world.

Algebra Term		ML Term
Slope (m)	→	**Weight (w)**
Intercept (c)	→	**Bias (b)**
y = mx + c	→	**y = wx + b**

Summary & Key Takeaways

- **Slope = Weight**: It determines how important an input feature is. A high weight means the input strongly affects the prediction.
- **Intercept = Bias**: It allows the model to shift its output, crucial for fitting data that doesn't pass through the origin.

Up Next

We will extend this concept from single numbers to lists of numbers using Linear Algebra (Vectors and Matrices).